XRPL Validator — Daily Engineering Report · 2026-09-30

Period: 2026-09-30 00:00 to 24:00 US Eastern (2026-09-30 04:00 to 2026-10-01 04:00 UTC); ledgers #107,329,222 to #107,351,502. Every count from the ledger and the validator's logs stops at #107,351,502. Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: the day's first and last ledger, from a public full-history server; signing records; Step 1 receipt records (they begin 2026-08-29); deployment records (they begin 2026-09-11); xrpl-validator merge records (they begin 2026-07-04); amendment data (carried from the original report). Not available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); Steps 2–5 receipt records (they begin 2026-10-09); data.xrpl.org agreement scores (scored days begin 2026-04-04; some days have none).

Status

Normal (agreement not checked) — coverage outside planned maintenance 100.000% (objective 99.9%), agreement with the network not checked (the reference node no longer holds these ledgers' hashes), no open engine receipt.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** signed 22,272 of the 22,281 ledgers the network closed (99.960%) (agreement not checked: the ledgers' hashes are no longer held). Unsigned: 9 during planned restarts.
- **Native engine:** 2 engine receipts recorded (2 fixed, 0 explained, 0 without a recorded fix).
- **Changes:** 1 deployment (cycle 162), 2 commits merged into main, 2 engine findings fixed (F409–F410).

Service levels

Measure	Value
Ledgers closed by the network	22,281
Validations issued	22,272
Coverage (validations / ledgers closed)	99.960%
Validations matching the network's validated ledger hash	not checked: the reference node no longer holds these ledgers' hashes
Unsigned during planned restarts	9
Unsigned outside maintenance	0
Coverage outside planned maintenance (objective 99.9%)	100.000%

Measure	Value
Restarts	1
Basis: the hash the first peer announced as accepted (before cycle 183)	22,272

Restarts:

- 03:34:16 UTC: deployment of cycle 162; validations resumed at #107,351,112 (9 ledgers unsigned between the last validation before the restart and the first after it)

The basis rows sum to validations issued. Coverage counts the validations the validator issued for ledgers the network closed in the period. Agreement compares each validation's ledger hash with the network's validated hash for that ledger. data.xrpl.org scores UTC days from the validations its crawlers received, so it can differ.

Native engine conformance

Ledgers with a receipt in the period (#107,329,222–#107,351,502), by step:

Step	Ledgers with a receipt
Step 1 · native engine	2
Step 2 · records	0
Step 3 · order	0
Step 4 · peer data	0
Step 4 · validations	0
Step 5 · own hash	0
Writer	0

Live counters are kept only for the current run; none exist for a past day.

Receipt register

ID	Time	Ledger(s)	Step	Description	Disposition
R-01	—	#107,330,165	Step 1 · native engine	3 ledger change(s) the network made were absent from ours: 30EECC9B, 20D424E0, A9D79B7A (deleted); 1 object(s) differed in content	fixed: F409, commit 32a6397, deployed in cycle 162 (2026-10-01 03:35 UTC)
R-02	—	#107,344,641	Step 1 · native engine	1 transaction result(s) differed from rippled's; 6 ledger change(s) the network made were absent from ours: 5E55169B, 9F8C163F, 8D2BD605	fixed: F410, commit 40c5fbd, deployed in cycle 163 (2026-10-02 03:47 UTC)

A receipt is recorded whenever the validator's own work differs from the network's, or the network's trusted validations reached it short of quorum. One ledger flagged by several steps is one receipt. Dispositions: fixed (with the finding, commit and deployment), explained (not a defect, with the reason), or open. Receipt and incident IDs are numbered per report: R-01 here is R-2026-09-30-01, INC-01 is INC-2026-09-30-01. Times are not shown where the reference node no longer holds the ledger.

Incidents

No incidents.

Change log

Deployments

Cycle	Main	Deployed	Test suites	Replay windows	Post-deploy health	Soak
162	`32a6397`	03:35:05 UTC	1,698 passed / 0 failed; port 745 / 0	HIST60 60 ledgers, 0 mismatches, WIN98 98 ledgers, 0 mismatches	404 matched, 0 mismatched	#37 from 04:01 UTC: ended early: superseded by the next deployment (2026-10-02 03:47 UTC)

Cycle 162 delivered: `32a6397` fix(xrpl-ledger): F409 — an AMM teardown deletes every trust line in the pool's directory

Each deployment passes the unit and port test suites and replays mainnet history windows with zero mismatches before it ships, is health-checked after it starts, and then soaks for 24 hours.

Merged into main

Merged	Repo	Commit	Subject
19:58 UTC	xrpl-validator	`32a6397`	fix(xrpl-ledger): F409 — an AMM teardown deletes every trust line in the pool's directory
02:38 UTC	xrpl-validator	`40c5fbd`	fix(xrpl-ledger): F410 — CheckCash judges DeliverMin on what the flow delivered, not on the casher's balance difference

Findings fixed

- **fix(xrpl-ledger): F409 — an AMM teardown deletes every trust line in the pool's directory** (`32a6397`, xrpl-validator) Mainnet #107330165 01DF0E1D0AE0: the last LP's tfWithdrawAll empties the BTC/BTC pool (rUnqjiCHbH3q), and rippled's deleteAMMAccount runs deleteAMMTrustLines (AMMUtils.cpp:237-280). That walks the pool's WHOLE owner directory through cleanupOnAccountDelete and deletes every trust line in it with deleteAMMTrustLine (View.cpp:3485-3528), not only the two asset lines. The directory still held another account's …
- **fix(xrpl-ledger): F410 — CheckCash judges DeliverMin on what the flow delivered, not on the casher's balance difference** (`40c5fbd`, xrpl-validator) Mainnet #107344641 48AADD252342: rw13qV1w cashes rKedDTMd's check for 1,034,740 "589" (DeliverMin = the check's SendMax = 1,034,740, no transfer fee) onto a trust line holding 112,178.5615725964. rippled's CashCheck compares DeliverMin with the flow's delivered amount (result.actualAmountOut), exactly 1,034,740, and mainnet cashes it (tesSUCCESS). The engine measured delivery as the casher's balance after minus …

Independence program

No independence record for UTC day 2026-09-30; records begin 2026-10-20.

Network

No ledger records for this day: the reference node holds ledgers from 2026-10-04 22:52 UTC (#107,436,647).

Amendment watch

Carried over unchanged from the original report, which read it from the XRP Ledger when it was generated; the reference node no longer holds this day's ledgers.

Read from the XRP Ledger: the Amendments object as of the last ledger of the day, and the day's EnableAmendment pseudo-transactions.

Amendment	Majority since	Activation due
PermissionDelegationV1_1	2026-09-24 21:25 UTC	2026-10-08 21:25 UTC
fixBatchV1_2	2026-09-25 14:12 UTC	2026-10-09 14:12 UTC
BatchV1_1	2026-09-25 14:46 UTC	2026-10-09 14:46 UTC

Amendments enabled on mainnet: 94.

Open items

None.

Corrections to earlier reports

None.

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **Forward-Priced NAV Orders for Tokenized Funds** (rwa) Add a `FundOrder` ledger object (`ltFUND_ORDER`, owner-reserved, `keylet::fundOrder(account, seq)`) and two transactors. `FundOrderCreate` (tfSubscrib
- **Rolling Spend Caps on Permission Delegations** (custody) Extend PermissionDelegation's `Delegate` ledger object with an optional `sfSpendLimits` STArray of up to 8 `SpendLimit` inner objects, each holding `s
- **Milestone-Tranche Escrow with Signature-Bound Release** (payments) Add an optional `sfTranches` STArray to `EscrowCreate` and the `Escrow` SLE: up to 8 `Tranche` inner objects, each with `sfAmount` and an optional `sf

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
09-30 19:58	09-30 19:58	xrpl-validator	`32a6397`	fix(xrpl-ledger): F409 — an AMM teardown deletes every trust line in the pool's directory
10-01 02:38	10-01 02:38	xrpl-validator	`40c5fbd`	fix(xrpl-ledger): F410 — CheckCash judges DeliverMin on what the flow delivered, not on the casher's balance difference

Times in UTC. A commit appears on the day it was committed to main or merged into it.